

Erasing Alpha

Davis, Han, Sokolinski, Tamoni

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- ▶ **Awesome!**
 - ▶ Arguments on *how* anomalies are eliminated often handwavy.
 - ▶ Provide explicit answer using i) existing estimates of price impact and ii) portfolio data

This Paper: Summary

The paper in one equation:

$$\text{Alpha-Erosion} \approx \underbrace{M}_{\text{Price Impact}} \times \underbrace{\theta}_{\text{Pass-Through to Exp. Returns}} \times \underbrace{h'g}_{\text{Alignment}}$$

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2. Quantitative magnitudes strongly hinge on M and θ .
Especially θ seems hard to pin down globally!

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This discussion: 1-3.

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(can be further decomposed as avg. stock-level illiquidity and strategy concentration)

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Compute capacity for strategies i . Test $R_{i,t} = \beta(\text{Post}_{i,t} \times \text{Capacity}_i) + \dots$

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(Btw: Eliminating an anomaly leads to initially high anomaly returns (Penasse, 2023))

Suggestion 3: The Value of Arbitrage

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- ▶ Does heterogeneity in capacity tell us more about the nature of the anomaly?
- ▶ A section on how to interpret the results would be awesome!
Below are some final thoughts
 - ▶ Capacity is primarily driven by stock size.
 - ▶ If mispricing exists in larger stocks the anomaly seems *economically* more important and perhaps even harder to create the first place.²
 - ▶ Corporate finance analogy: NPV versus IRR.
 - ▶ Davila, Graves, Parlato (JPE, 2024) link the amount of capital required to eliminate arbitrage opportunities to the welfare cost of the arbitrage.

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Summary

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- ▶ 4 suggestions that would perhaps increase the likelihood of people adopting the measure/framework

1. Reshift focus away from *grand* counterfactuals.
2. Simplify **capacity** measure. What drives heterogeneity across strategies?
3. Provide empirical evidence on its usefulness.
4. More discussion on what this means for the nature of different anomalies.

Literature

1. **Disappearance/decay of anomalies:** *McLean and Pontiff (2016)*

Chordia, Subrahmanyam, Tong (2014); Green, Hand, Zhang (2017); Harvey, Liu, Zhu (2016); Chen and Zimmermann (2022); Jensen, Kelly, Pedersen (2023); Falck, Rej, Thesmar (2022).

2. **Trading costs of anomalies:** *Novy-Marx and Velikov (2016)*

Korajczyk and Sadka (2004); Lesmond, Schill, Zhou (2004); Frazzini, Israel, Moskowitz (2018); Detzel, Novy-Marx, Velikov (2023); Patton and Weller (2020); Chen and Velikov (2023).

Suggestion 1b: Define Capacity (Volume Weights)

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- Capacity is a unique feature of a strategy. Can compare strategies!
(can be further decomposed as avg. stock-level illiquidity and strategy concentration)
- Hard to erase alpha: Total Stock Market Cap is > \$50 Trillion!